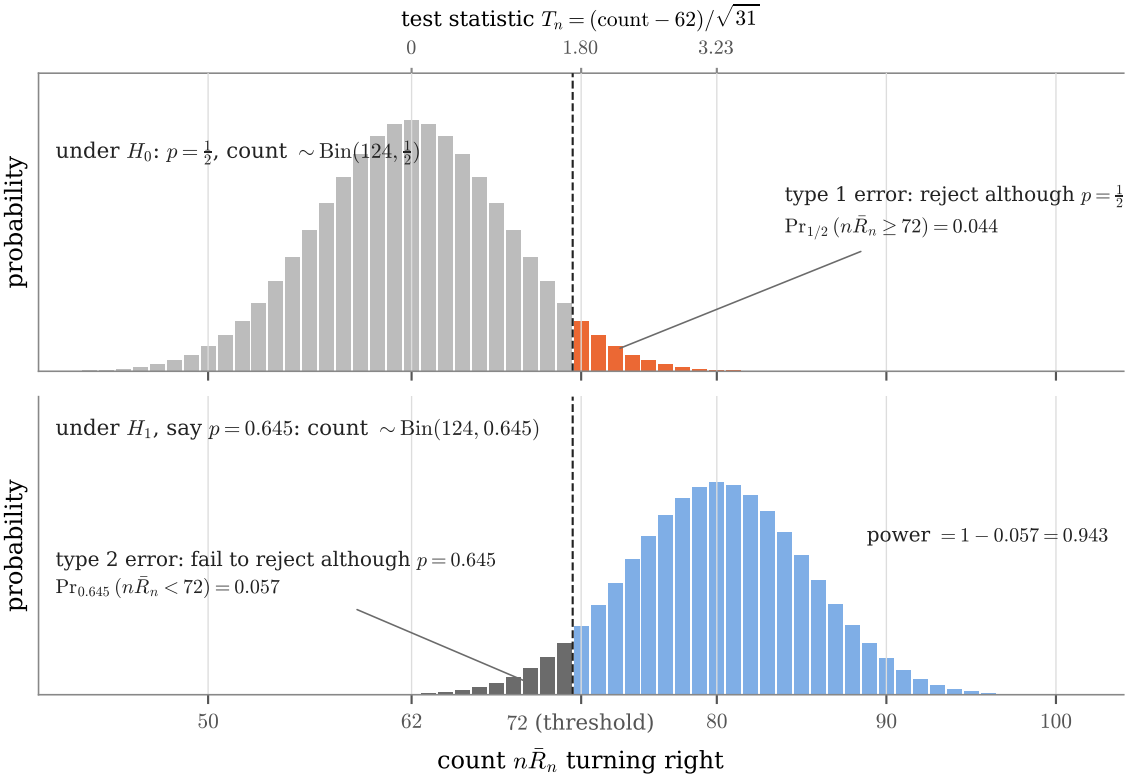
test statistic $T_n = (\text{count} - 62)/\sqrt{31}$
0
1.80
3.23
probability
under H_0: $p = \frac{1}{2}$, count $\sim \text{Bin}(124, \frac{1}{2})$
type 1 error: reject although $p = \frac{1}{2}$
$\Pr_{1/2}(n\bar{R}_n \geq 72) = 0.044$
probability
under H_1, say $p = 0.645$: count $\sim \text{Bin}(124, 0.645)$
type 2 error: fail to reject although $p = 0.645$
$\Pr_{0.645}(n\bar{R}_n < 72) = 0.057$
power $= 1 - 0.057 = 0.943$
50
62
72 (threshold)
80
90
100
count $n\bar{R}_n$ turning right